

Fishing Rod Holder Teacher Guide

Year 9 Metalwork | Answer guidance for the 50-page student textbook

AUTHORITY Use the current teacher-issued plan, brief, demonstrations, SOPs and approved assessment notice. The original components drawing is NOT TO SCALE. The 2026 assessment draft requires approval; the separate 2027 program has a different syllabus context. Do not treat this guide as a machine permission or marking schedule.

Use the student page numbers below to locate each activity. Model responses are examples; open tasks accept other well-supported reasoning. Practical observations must reflect actual student work.

Page 04 Risk management is a process, not a poster

Student task: Choose one stage shown in your teacher-issued project instructions. Record (1) the stage and observed hazard, (2) how likely harm is and why, (3) the possible consequence, (4) one stronger control and any directed PPE, and (5) a change that would make you stop and review. Write two sentences explaining why your stronger control matters more than PPE alone.

Look for: A real teacher-approved stage; a specific hazard rather than a generic warning; reasoned likelihood and consequence; a suitable stronger control checked against local practice; directed PPE as an additional layer; and a clear stop-and-review trigger.

Example reasoning: If stock is not secure at an approved workstation, it may move unexpectedly and cause injury. The setup must be checked and secured using the teacher-approved method before work begins; PPE alone cannot keep the stock in place.

Acceptable alternatives: Students may choose another approved project stage and different valid controls. Do not assess a named machine method as authorised until local induction and teacher direction are confirmed.

Page 05 Safe workshop practices

Student task: Create a two-column “ready / needs teacher check” record for an upcoming approved project stage. List at least three conditions you can check yourself and three that require current teacher confirmation. Explain what you will do if a guard, setup or work area differs from the demonstration.

Look for: Self-checks may include attention, clothing/hair, clear bench and walkway, and storage. Teacher checks include machine permission, demonstrated setup, local SOP, exact PPE and supervision. A changed setup leads to stop and seek teacher direction.

Example reasoning: I can clear offcuts from the walkway and secure loose clothing. I need the teacher to confirm that this machine and setup are approved for this task. If a guard is different, I stop and ask before work begins.

Acceptable alternatives: A condition can move between columns if the student explains the local teacher-controlled boundary; safety authority remains with the teacher.

Page 06 Incident response and reporting

Student task: Use a teacher-provided scenario or the hypothetical shifted-stock example. Write four brief entries: what you observed; what you did immediately; what you must tell the teacher; and which control needs review. Underline any statement that is only a guess and rewrite it as a question to check.

Look for: Stops work, keeps the area safe, alerts the teacher, reports observable facts and identifies a control for review. Guesses are labelled as uncertain. No unsupported student-led restart or repair.

Example reasoning: Observed: the stock shifted before anyone was hurt. Immediate action: stop and keep others clear. Tell the teacher the task and setup. Ask whether workholding was correct rather than claiming the vice failed.

Acceptable alternatives: Teacher-provided incidents may require different immediate actions under the school emergency procedure. A student should not be marked against an invented local reporting form.

Page 07 Check your understanding 1

M1-Q01 A — Risk level depends on both the chance of harm and how serious the possible harm could be.

M1-Q02 B — Existing controls may no longer be effective when the work area, equipment, material or people nearby change.

Source written prompt: Choose one stage from the teacher-issued project plans. Identify one possible hazard, describe its likelihood and consequence, and explain which hierarchy-of-controls measures should be used before work begins.

Example feedback: During drilling, an unsecured workpiece could spin. This is possible and could cause a serious hand injury. I would use the approved vice or clamp and machine guard, follow the teacher's setup and wear the directed PPE. If anything changes, I would stop and seek teacher direction.

Page 08 Risk control record

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 03 — WHS and risk controls: Record hazards, likelihood, consequence, controls, PPE and permission points.

Teacher confirmation: Teacher to confirm the actual stage and approved controls before practical work.

Page 09 Choose metal for the job

Student task: Choose one component on the drawing. In separate columns record its exact printed material/stock label, one relevant property to investigate, how that property could affect performance, and the source or teacher confirmation needed. Finish with a two-sentence material justification that makes no unsupported claim about grade or finish.

Look for: An exact drawing label for the selected component, a relevant property connected to its function, and a clear distinction between printed fact and proposed performance. Grade, coating and process claims are checked with the teacher.

Example reasoning: The plan specifies Ø12 black bar for the stake. I would investigate whether the stock has enough strength and stiffness for the approved use and how it will be protected from corrosion. The teacher must confirm the material grade and finish before I claim these details.

Acceptable alternatives: Rod holder or rotation tab labels are equally valid. Different properties may be defensible when tied to the approved brief and evidence.

Page 10 Properties become performance

Student task: Choose one property and explain how it could change a material decision for an approved project part.

Look for: A property is linked to function and a justified choice; no unapproved material substitution.

Page 11 The Fishing Rod Holder brief and production context

Student task: Read the current teacher-issued brief. Copy two actual requirements and one actual constraint into a three-row table. For each, write how it could be checked and what evidence you could save. Add one preference of your own and explain whether it is approved. If the brief is unavailable, mark the relevant cells "Teacher to confirm".

Look for: Verbatim or accurately paraphrased teacher-issued requirements and constraint, sensible check/evidence pairings, and an explicit distinction between preference and required criterion.

Example reasoning: For a teacher-confirmed drawing requirement, the student could record the printed dimension used, their measured result and a caption explaining the comparison. The teacher must confirm the actual criterion and acceptable tolerance.

Acceptable alternatives: The brief may specify different requirements. If unavailable, "Teacher to confirm" is correct; do not reward fabricated limits or an unapproved test.

Page 12 Design and project planning

Student task: Draw a five-box planning flow from teacher-issued brief to approved practical work. Include the working drawing, cutting list, a quality checkpoint and a teacher permission point. Under each box, state what evidence you would save. Leave the exact operation and machine setting blank until demonstrated and approved.

Look for: A logical relationship among brief, drawing, cutting list, approved sequence, checkpoint and permission; saved evidence attached to decisions; no invented operation, dimension or machine setting.

Example reasoning: Brief → read component drawing → record checked cutting list → ask teacher to confirm the first operation and setup → carry out the approved check before proceeding. A revision note states what evidence showed the plan needed changing.

Acceptable alternatives: The teacher may approve a different order of planning artefacts; the actual workshop operation order is not inferred from this generic flow.

Page 13 Read the Fishing Rod Holder components drawing

Student task: Using the unchanged teacher-issued drawing, annotate the title, “NOT TO SCALE” note and the views for each of the three named parts. Complete a register with each printed part name, stock label and quantity. Copy one printed dimension or note for each part exactly as it appears, and write one question for the teacher about any uncertain unit, relationship or practical step. Do not take measurements from the page.

Look for: Correct title; explicit “NOT TO SCALE”; ROD HOLDER, STAKE and ROTATION TAB matched to their own views; labels 50 nb. pipe, Ø12 black bar, 25 × 5 flat; one of each. Dimensions/notes copied from the actual plan, not measured from the print.

Example reasoning: ROD HOLDER: 50 nb. PIPE, 1 OFF. STAKE: Ø12 BLACK BAR, 1 OFF. ROTATION TAB: 25 × 5 FLAT, 1 OFF. The rod holder length is printed as 100; the stake is labelled “300 ORIGINAL LENGTH BEFORE FORGING”; the rotation tab is shown with printed 50 and Ø8 dimensions. A teacher should confirm any unit or manufacturing interpretation that is unclear.

Acceptable alternatives: Other exact printed dimensions or notes for the same component are acceptable. Do not convert a plan note into an unsupervised instruction for forging, drilling, tapping, machining or joining.

Page 15 Read the rod holder views

Student task: Annotate the ROD HOLDER views here, then complete the plan-reading record on page 13. Identify the printed material, quantity and one feature note. Explain why no size can be taken with a ruler from this page.

Page 16 Read the stake and rotation tab

Student task: Create a two-row component record. For each part, copy its exact material and quantity wording, identify one feature or note, and write one question you would check before making.

Page 17 Check your understanding 2

M2-Q01 A — Properties describe characteristics while performance describes how effectively the metal works in context.

M2-Q02 C — A sound justification connects relevant properties to the full project context.

M2-Q03 A — The verified drawing title identifies the sheet as the component drawing for the Fishing Rod Holder.

M2-Q04 B — The drawing names the ROD HOLDER, STAKE and ROTATION TAB as the three project components.

Page 18 Read and reason 2.2

M2-Q05 C — The sheet is not to scale, so only its printed dimensions and notes provide valid drawing information.

M2-Q06 B — The quantity label shows that one of the named components is required for the project.

M2-Q07 C — The verified component information identifies the STAKE as Ø12 black bar, with one required.

M2-Q08 A — Several forms of printed information should be traced together before deciding which views represent each part.

Page 19 Read and reason 2.3

M2-Q09 C — The drawing identifies the ROTATION TAB as 25 x 5 flat and specifies a quantity of one.

M2-Q10 D — Uncertainty must be resolved through the teacher because the issued information controls the authorised project requirements.

M2-Q11 C — Saved evidence should show that component identities, quantities, material labels, dimensions and notes were checked.

M2-Q12 B — Teacher-issued plans, demonstrations and instructions control settings, sequence, materials, dimensions and permissions.

Source written prompt: Using the teacher-issued plans, identify the selected metal for one project part and justify the choice by referring to function, fabrication, corrosion, availability and finish.

Example feedback: The teacher-issued plan specifies [metal] for [part]. Its [relevant property] suits the part's function, and it can be fabricated using the approved workshop process. I would also explain how the material suits the expected corrosion conditions, availability and finish.

Page 20 Brief and constraints

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 01 — Project brief: Summarise the need, user and function for the teacher-issued Fishing Rod Holder brief.

Folio 02 — Requirements and constraints: Separate essential requirements, preferences and constraints for the Fishing Rod Holder.

Teacher confirmation: Use the teacher-issued brief; do not invent formal task conditions.

Page 21 Materials and ideas

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 04 — Materials investigation: Compare relevant properties and justify teacher-specified material choices.

Folio 05 — Research and ideas: Record useful research, concept evidence and reasons for the approved direction.

Page 22 Working drawing evidence

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 06 — Working drawing evidence: Refer to teacher-issued drawings and explain the information used for production.

Teacher confirmation: Use the original drawing and leave the sketch field clear for your annotations.

Page 23 Accuracy before force

Student task: Choose a teacher-approved component or practice shape. Make a quick concept sketch showing the demonstrated reference, one printed dimension or note, material to keep and waste side where relevant. Add four pre-operation checks and a question for anything unclear. Develop the larger datum example on page 26.

Look for: One consistent demonstrated reference, correct orientation to printed information, keep/waste distinction where applicable, secure workholding as a check and a pause when source information is unclear.

Example reasoning: I will mark from the teacher-demonstrated reference and compare with the printed dimension. I will check orientation and waste side, then confirm the stock is held as demonstrated.

Acceptable alternatives: Round stock or some operations may not have a datum edge or conventional waste side. Accept a teacher-approved reference and justified “not applicable”; do not force the flat-stock example onto the holder.

Page 24 Marking and measuring tools

Student task: For four items of teacher-approved drawing information, record the feature, proposed tool, what it checks or marks, and how you will recheck it. Mark any centre-punch or machine-dependent step “Teacher to confirm”. Explain why a tidy line can still be inaccurate.

Look for: Rule/tape for suitable length, square for perpendicular reference, scribe for clear line and centre punch only for an approved hole centre; each choice is tied to source information and rechecked.

Example reasoning: A tidy line may be wrong because it was measured from the wrong reference, copied from the wrong view or marked on the wrong side.

Acceptable alternatives: Different school-approved tools may suit a component. Judge reasoning against the actual teacher demonstration.

Page 25 Cutting and forming sequence

Student task: For a teacher-demonstrated operation, record three stages in the table: check before, approved change to stock, check after. Name one evidence item at each stage and one possible problem if the order changed. Mark any unconfirmed exact process or allowance ‘Teacher to confirm’.

Look for: An authorised operation with a meaningful source and setup check, post-operation observation, and an explanation of how sequence protects fit, access, size or safety.

Example reasoning: Before the approved operation I check the plan, marks, orientation and secure setup. Afterwards I record the measured or observed result. A changed order might block access, so I ask the teacher first.

Acceptable alternatives: Actual operation and order vary. If the stage has not been taught, a conceptual map with “Teacher to confirm” is appropriate.

Page 26 A datum prevents drift

Student task: On an imagined rectangular offcut, mark one datum edge, two measurements from that edge, the waste side and the point where you would recheck. This is a conceptual marking exercise, not a project cutting plan.

Look for: One consistent reference edge, correctly distinguished waste side, and a check before processing. Specific project dimensions require the current plan.

Page 27 Check your understanding 3

M3-Q01 A — A datum gives measurements a consistent starting point and reduces accumulated errors.

M3-Q02 A — Securing the work and checking datum, orientation, waste side and sequence support safe, accurate processing.

Source written prompt: Explain how a datum edge, waste-side marking, planned sequence and secure workholding can improve the quality of a sheet-metal project.

Context for this prompt: The source question refers to a sheet-metal project. Treat it as a hypothetical sequence question; the Fishing Rod Holder plan specifies pipe, bar and flat stock.

Example feedback: I measure from one datum edge so each mark has the same reliable starting point. I mark the waste side to protect the finished size, follow the planned sequence so later features still line up, and secure the work so it cannot move while I cut or form it.

Page 28 Cutting list and resources

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 07 — Cutting lists and resources: Record teacher-specified parts, materials, tools and consumables without guessing.

Teacher confirmation: Teacher to confirm current dimensions, resources and permissions.

Page 29 Production plan and checkpoints

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 08 — Production planning: Sequence work, checkpoints, demonstrations and approvals for the Fishing Rod Holder.

Teacher confirmation: Record the sequence actually directed by your teacher.

Page 30 Production evidence 1

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 09 — Production evidence: Add captions for marking, cutting, forming, joining, machining and finishing stages.

Page 31 A joint is a design decision

Student task: For one teacher-identified connection, make six rows for load, permanence, access, heat effect, appearance and corrosion. State what each factor asks you to check and which facts remain teacher controlled. Add the teacher-requested trial-fit observations before any permanent joining. Do not nominate an approved method unless one has been issued.

Look for: Each factor is connected to a practical consequence; the actual method comes from teacher authority; a trial fit checks contact, alignment and position before permanence.

Example reasoning: A method requiring tool access may not suit a hidden connection. A permanent joint makes fit errors harder to correct. The teacher must confirm the method and trial-fit criteria.

Acceptable alternatives: Different authorised methods can be valid. The source rivet video does not prove the holder is riveted.

Page 32 Choose a joining method

Student task: Choose two decision factors and explain how each could change a joining decision. Do not nominate an unapproved process as the project method.

Look for: Two factors linked to functional or manufacturing evidence; approval boundary retained.

Page 33 Machining is permission-based

Student task: Copy one machine-related note exactly from the plan. In two columns list what the drawing tells you and what it does not authorise you to decide. Add a “stop and report” statement for changed sound, vibration or stock position. Ask the teacher for current local confirmation; the paper task is not machine permission.

Look for: An exact drawing note, distinction between specification and operating authority, and safe stop-and-alert action for changed conditions.

Example reasoning: The note identifies a required feature. It does not say I can use the machine or choose its speed. If the work moves or sounds different, I stop and tell the teacher.

Acceptable alternatives: The teacher may select another note or perform the operation. A completed page is not student machine sign-off.

Page 34 Surface preparation and protection

Student task: For the teacher-approved finish, record the product name from its label, surface condition before preparation, demonstrated preparation, one safety control from the actual product or teacher instruction, and the result after curing. Add a real photo caption or labelled sketch. If no finish is issued, mark product and process “Teacher to confirm” and complete the concept explanation.

Look for: Actual product and process from teacher/product information; surface condition linked to coating quality or corrosion protection; a specific observation; no invented ventilation, PPE, coat count or cure time.

Example reasoning: The approved preparation removed contamination noted at inspection. My caption explains what the result shows about coverage and any visible defect.

Acceptable alternatives: Different products and processes may be approved. “Teacher to confirm” is correct when none has been issued.

Page 35 Trial fit before permanence

Student task: Write a hypothetical three-line trial-fit record: the feature checked, evidence seen, and the decision or teacher question that follows. Keep it separate from your real production record.

Look for: Specific observation and justified next action; no invented actual project result.

Page 36 Check your understanding 4

M4-Q01 A — A suitable joint must match functional, manufacturing and environmental needs.

M4-Q02 A — A trial fit reveals contact, alignment and position problems while they can still be corrected.

Source written prompt: Explain how load, permanence, access, heat effect, appearance and corrosion could influence a joining method. Include why a trial fit is needed.

Example feedback: The joining method must carry the expected load and suit whether the part needs to come apart. I would also check access, heat effects, appearance and corrosion. Before permanent joining, I would trial fit the parts to check contact, alignment and position.

Page 37 Production evidence 2

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 09 — Production evidence: Add captions for marking, cutting, forming, joining, machining and finishing stages.

Page 38 Quality checks

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 10 — Quality checks: Record measurements, trial fits, alignment checks and teacher-approved criteria.

Teacher confirmation: Use only criteria and tolerances supplied or approved by the teacher.

Page 39 Problem solving and testing

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 11 — Problem solving and testing: Explain a problem, evidence, decision, correction and safe test result.

Page 40 Quality, technology and evaluation

Student task: Create three rows for real teacher-approved checkpoints: criterion, evidence, what it proves or cannot prove, and decision. Choose one row and write an evidence-supported judgement plus one realistic improvement. If the stage has not happened, label the result “Awaiting observation”.

Look for: Relevant checkpoints tied to current criteria, honest limits for each evidence type, and a specific judgement and improvement. A photo alone does not prove teacher-observed practical skill.

Example reasoning: A trial-fit image can show visible alignment but cannot alone prove a printed size. Pair it with a teacher-approved measurement record and seek direction before permanent joining if fit is wrong.

Acceptable alternatives: Different authentic checkpoints are valid. “Awaiting observation” is honest before a practical stage.

Page 41 Current and emerging manufacturing

Student task: Compare one generic metal practice part with a batch of identical parts. For each, judge conventional and CNC or another emerging process against accuracy, quantity, material, resources/setup and safety authority. Choose a better-supported option for each scenario and name one condition that could change the decision. Keep actual Fishing Rod Holder production teacher controlled.

Look for: Conditional comparison against named criteria, not “newer is better”. Conventional may suit a flexible one-off; a validated digital process may suit a repeat batch if resources and trained operators exist.

Example reasoning: For one part I may choose an approved conventional process because digital setup is not justified. For a batch, a validated CNC process might improve repeatability if a machine, file, suitable stock and trained operator are available.

Acceptable alternatives: An emerging process can be selected for either scenario when the student offers sound task-specific evidence. The comparison authorises no method for the holder.

Page 42 Conventional and CNC choices

Student task: Choose a hypothetical component and compare two process options against three criteria. Explain your choice as a reasoned example, not an instruction for this project's workshop.

Look for: A fair comparison using shared criteria; no claim that newer is always better.

Page 43 Folio and final evaluation

Student task: Choose three authentic pieces of your own evidence from planning, production and final checking. Caption each with the decision or approved criterion it supports and one limit of that evidence. Then write an evaluation with one evidenced strength, one evidenced limitation and one realistic improvement. Label an incomplete panel “Awaiting evidence”.

Look for: Chronological authentic evidence; captions explaining what each item supports and cannot prove; evaluation against actual criteria with a feasible improvement. No invented observations or automatic claim of practical competence.

Example reasoning: My planning record shows the drawing notes used. A teacher-observed trial-fit note shows how alignment was checked. My final photograph shows appearance. I identify one documented limitation and explain how an earlier check might improve it.

Acceptable alternatives: Sketches, measured records, teacher observations and photos can all be valid. “Awaiting evidence” is correct until a stage is complete.

Page 44 Check your understanding 5

M5-Q01 B — Suitability depends on accuracy, quantity, material, resources and purpose.

M5-Q02 C — A strong evaluation names evidence, explains the result and identifies a realistic improvement.

Source written prompt: Compare one conventional process with one CNC or emerging process by explaining when each would be suitable. Then evaluate one part of your project using specific evidence and identify one realistic improvement.

Example feedback: Hand cutting can suit a one-off part because it is flexible and develops practical skill. CNC can suit a batch when repeatability and accurate digital information matter. My [measurement, trial fit or photo] showed _____. Next time I would _____ so the result better meets the approved criteria.

Page 45 Final evaluation

Student task: Write a connected evaluation. State the approved criterion, cite your actual evidence, explain the judgement and propose one realistic improvement.

Look for: Specific, truthful evidence for the named folio item, linked to the teacher-issued brief and actual project work. Check local practical and assessment requirements before sign-off.

Folio 12 — Final evaluation: Judge function, accuracy, appearance and construction quality with evidence and one realistic improvement.

Page 46 Improve one feature

Student task: Draw an annotated improvement concept. Label the feature, the evidence that prompted the change and one likely trade-off.

Look for: A labelled proposal grounded in a real observation and a plausible trade-off; not a claim that an unapproved change was built.

Page 47 Make your folio tell the story

Student task: For each folio section, mark evidence present, evidence still needed or teacher confirmation. Then write the next two actions that would most improve the clarity of your folio.

Page 48 Explain a drawing in views

Student task: Label each view, add centre or hidden lines only where needed, and explain which written note would be essential before manufacturing.

Look for: Complementary labelled views and an explanation of source-backed dimensions/notes; no inferred project measurement.

Page 50 Check the evidence, then decide

Student task: Write one sentence each: a decision you can justify, a quality check that changed your work, and a question to take to your teacher.

Source and media record

Course: <https://stevenowell.github.io/Yr-9-Metal/fishing-rod-holder/> (repository commit 949f90eb6bc28bcd010fb43b50f27ae956ee1c9). Original plan: FISHING ROD HOLDER: COMPONENTS, source SHA-256 0697881507f41a4326d2cb6f2eee49afe5f40a5ea5f118c6c46b4cc453707713. Folio: course folio.html.

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